

URL Shortener – Proof of Concept (PoC)

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Project: URL Shortener using Node.js + SQLite + nanoid

Objective

To design a simple web-based URL Shortener that allows users to input a long URL and receive a unique short version for easy sharing and redirection.

Problem Statement (Simplified)

Long URLs are often hard to share, remember, or type — especially on social media or messaging platforms where space is limited. People want a way to make these links shorter, easier to manage, and more user-friendly.

The goal of this project is to build a simple tool that lets users:

- Enter a long URL
- Get a short version of that URL
- Click the short link and be redirected to the original one
- Record when the short link is used (for basic tracking)

This tool should be easy to use, run locally, and store data safely using a small database.

Tools Used

- **Node.js** – Backend runtime
- **Express.js** – Web framework
- **SQLite3** – Lightweight local database

- **nanoid** – Generates unique 6–8 character short slugs
 - **EJS** – Template engine for rendering views
 - **HTML** – Frontend
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Project Structure

```
pgsql

url-shortner/
├─ public/
│   └─ index.html
├─ views/
│   └─ result.ejs
├─ server.js
├─ package.json
└─ urls.db
```

Use Case

A user wants to share a long URL on social media, but the link is too long and looks messy.
The user opens the URL Shortener tool, pastes the long URL into a form, and clicks "Shorten."
The tool gives a short, clean link that redirects to the original URL when clicked.
The tool also tracks who clicked the short link and when.

Setup and Installation

Frontend Form:

html

```
<form action="/shorten" method="POST">
  <input type="text" name="longUrl" placeholder="Enter long URL" required />
  <button type="submit">Shorten</button>
</form>
```

Backend Installation:

bash

```
npm init -y
npm install express body-parser sqlite3 ejs nanoid
```

- Initializes a Node.js project
- Installs required libraries:
 - **express:** web server
 - **body-parser:** handles form input
 - **sqlite3:** database to store URLs
 - **ejs:** template rendering
 - **nanoid:** generates secure 7-character short codes
- Create **server.js**
- Create HTML: **public/index.html**

- Create result view: `views/result.ejs`

Database Setup

```
const sqlite3 = require('sqlite3').verbose();
const db = new sqlite3.Database('./urls.db');

db.serialize(() => {
  db.run('CREATE TABLE IF NOT EXISTS urls (id TEXT PRIMARY KEY, longUrl TEXT)');
  db.run('CREATE TABLE IF NOT EXISTS clicks (id INTEGER PRIMARY KEY
AUTOINCREMENT, shortId TEXT, timestamp TEXT, ip TEXT)');
});
```

Short Code Logic with nanoid

```
js

const { nanoid } = require('nanoid');
const id = nanoid(7);
```

- Secure
- URL-safe
- Randomly generated

Shorten Route Logic

```
js

app.post('/shorten', (req, res) => {
  const longUrl = req.body.longUrl;
  const id = nanoid(7);

  db.run('INSERT INTO urls (id, longUrl) VALUES (?, ?)', [id, longUrl], (err) => {
    if (err) return res.send('DB Error');
    res.render('result', { shortUrl: `http://localhost:${PORT}/${id}` });
  });
});
```

Result View (EJS)

```
ejs

<h2>Your Short URL</h2>
<a href="<%= shortUrl %>"><%= shortUrl %></a>
```

Redirection Logic

```
js Copy Edit

app.get('/:id', (req, res) => {
  const id = req.params.id;

  db.get('SELECT longUrl FROM urls WHERE id = ?', [id], (err, row) => {
    if (err || !row) return res.send('URL not found!');

    const timestamp = new Date().toISOString();
    const ip = req.ip;

    db.run('INSERT INTO clicks (shortId, timestamp, ip) VALUES (?, ?, ?)', [id, timestamp, ip]);
    res.redirect(row.longUrl);
  });
});
```

User Flow

1. Visit Tool: Open <http://localhost:3000>
 2. Enter URL: Enter a long URL
 3. Click Submit: Click the “Shorten” button
 4. Receive Short URL
 5. Share Link
 6. Click Tracking: Time and IP logged
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Test Cases

Test Case	Expected Output
Valid long URL	Short URL generated
Invalid URL	Error message
Visit valid short URL	Redirect to original URL
Visit invalid short code	Show "URL not found"

Benefits of the Tool

- Simplifies Link Sharing
- Secure & Unique (nanoid)
- Stores Data Locally with SQLite
- Easy to Maintain
- Ready for Deployment

Who Can Use the Tool?

- **General Users:** Share clean links
- **Students & Developers:** Learn web routing and DB
- **Marketers:** Share and track campaigns
- **Organizations:** Internal URL management
- **Cybersecurity Interns:** Understand redirection misuse

Conclusion

This URL Shortener project demonstrates:

- **Full-stack development (Node.js + Express + SQLite)**
- **URL redirection and tracking with short codes**
- **Secure random ID generation using nanoid**
- **Real-time redirection and visit logging**